Jialin Li

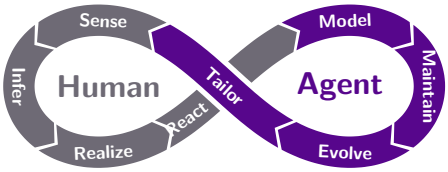
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Education

New York University — Sept. 2024 – May 2029 (Expected)
Ph.D. in Computer Science, NYU Abu Dhabi Global Ph.D. Fellow — *New York, NY, USA*
Advisor: Hanan Salam

New York University Shanghai — Sept. 2019 – Jan. 2023
B.S. in Data Science — *Shanghai, China*

Research Interests



Research Focus: People-Aware LLM Agents. I build and benchmark personal agents that adapt to individual personality, context-dependent interaction preferences, and continuous user feedback.

Long-Term Goal: Co-evolving human-agent partnerships, turning daily interactions into learning signals that adapt agent harnesses to amplify human agency without constraining it.

① Human Signals
Q: How to *sense* and *model* what makes each person distinct?
How: Personality-aware & per-user multimodal affect modeling
MOCSE · PersonAug · PAC Survey

→

② Preference States
Q: How can agents *infer*, *maintain*, and *realize* how you prefer to interact?
How: Context-resolved preference adaptation & long-horizon memory
PrefIx · MemPABench

→

③ Adaptive Harnesses
Q: How can models and harnesses *co-evolve* from users' feedback?
How: Turning users' interactions into updates for alignment & harness
Alignment & Harness Evolution

Experience

Tencent Hunyuan, Foundation Model Team — May 2026 – Sept. 2026
Algorithm Research Intern, In-Game Companion Agents — *China*

- **Interactive Data Synthesis & Evaluation ③:** To overcome scarce state-grounded dialogue, context drift, and unnatural TTS cadence, developed a **two-sided in-game synthesis pipeline** integrating **player personas** and NPC **cognitive profiles** with state-alignment rules, synthesizing **10K+** role-consistent multi-turn dialogues. Implemented a two-layer rubric evaluator applying **gating metrics** (safety, logic) as hard filters to discard invalid samples and **uplift metrics** (spoken style, plot progression) for fine-grained ranking. Calibrated the evaluator against human ratings to standardize data filtering and RL reward modeling.
- **Real-Time State & Layered Memory ②:** Rapid PUBG state changes and **asynchronous** reasoning often led to stale context and module-level hallucinations. Consolidated game state, match events, and dialogue into a shared, **incrementally updated structured context**, cutting mismatched triggers by **8%**. Designed a lifecycle-based **L0–L4 layered memory** framework with automatic summarization, constructing targeted training data to optimize memory recall and natural follow-up generation.
- **Implicit-Feedback Reward & Multi-Turn RL ③:** Single-turn RL failed to optimize long-horizon conversational goals over multi-turn dialogues. Modeled implicit player preferences across **behavioral signals** (mic usage, chat continuity) and **subjective states** (sentiment, interaction tone) to calibrate a **multi-objective reward model**. Executed staged training: SFT for prompt compression and persona style, followed by **offline PPO and online GRPO** for boosting composite reward scores.
- **Attribution & Data Flywheel ③:** Built an online bad-case analysis pipeline processing **20K** live sessions a day. Replaced single-point scoring with multi-stage analysis to isolate root causes by **system layer**, cutting regression evaluation cycles by **50%+**. Closed a **feedback → attribution → retraining data flywheel** for targeted data iteration, supporting a product serving **8M total users** with a peak DAU of **500K**.

Publications

ACL Findings 2026 **PrefIx: Understand and Adapt to User Preference in Human-Agent Interaction**
J. Li, Z. Chen, H. Luo, H. Salam. Code ②

- **Motivation:** LLM agents nowadays can solve tasks correctly, but their interaction still frustrates users in various ways, e.g. excessive confirmations. Existing benchmarks evaluate accuracy alone, missing the interaction flaws.
- **Approach:** Introduced ***interaction preferences*** and **Interaction-as-a-Tool (IaaT)**. IaaT frames models interaction behaviors (confirm, disambiguate, escalate etc.) as structured tool calls, to make them evaluable with standard tool-use benchmarks. We test this across 31 preference in four dimensions (transparency, interaction pace, strategic proactiveness, robustness) with simulated users. Evaluated on seven UX dimensions by a panel of four LLM judges.
- **Findings:** Our agents match baseline accuracy while improving UX by 7.6% and alignment by 18.5%. Among the four preference dimensions, transparency was the easiest to satisfy, while robustness was the hardest.

Under Review **Why Bother Me Again? Benchmarking Personal Agent Memory for Interaction Preferences**
J. Li, Z. Chen, H. Luo, H. Salam. Code Webpage ②

- **Motivation:** In long-term assistance, personal agents must track changing user preferences across contexts and over time to keep interactions satisfying. However, current memory benchmarks fail to evaluate these evolving preferences.
- **Approach:** Built an interactive environment for **on-policy memory evaluation MemPABench**. Presented four grounded simulated users with 28 interaction preferences across work and personal contexts. They used the testing agents over 99 sessions across four months, including five preference shifts. Evaluated four memory systems across various LLM backbones on preference selection, delivered behavior, and user-perceived interaction quality.
- **Findings:** Retrieval-based memory (Mem0, MemOS, RAG) merely matches the no-memory baseline. File memory performs better, but it overgeneralizes in misapplying preferences in the wrong contexts. Worse, agents struggle with execution. Even given the right preference, they deliver the intended behavior only 59% of the time.

ACII 2025 (Oral) **Improving Personalisation in Valence and Arousal Prediction using Data Augmentation**
J. Li, M. Nwadike, H. Salam. ①

- **Motivation:** Per-person valence and arousal models have little data for each individual.
- **Approach:** Proposed a data-driven **sample-level augmentation** that pulls similar segments from multi-person data to adapt the target model. Improved CCC on personalization benchmark without changing model architectures.

ACM MM 2025 Workshop **Personality-Aware Engagement Prediction in Online Learning**
J. Li, G. Sharma, H. Salam. ①

- **Motivation:** Vision-based engagement prediction ignores individual differences, and existing datasets are mostly lab-collected, one-dimensional, and lack self-reports and personality.
- **Approach:** Built MOCSE, an in-the-wild dataset of Zoom lectures (72 students, 36.5 hours) with self-reported and observer engagement labels plus Big Five personality; conditioning VLMs on personality improves prediction.

arXiv 2023 **Beyond One-Size-Fits-All: A Survey of Personalized Affective Computing in Human-Agent Interaction**
J. Li, M. Elgarf, A. Waleed, H. Salam. ①

- First survey of personalized affective computing, with a taxonomy of data- and model-level approaches.

Skills

Evaluation & Data: Interactive benchmark and environment design, LLM-as-a-judge, multi-turn data synthesis
Post-Training: SFT, reward modeling, PPO, GRPO
Programming: Python, Rust, Shell, SQL; PyTorch, Transformers; Docker, Git, Linux, WandB; Claude Code, Codex

Honors & Awards

NYU Abu Dhabi Global Ph.D. Fellowship, New York University 2024 –
University Honors Scholar, New York University 2023
Dean's List of Academic Year, New York University 2021 – 2022
Dean's Undergraduate Research Fund, New York University 2020 – 2021

Service

Reviewer: ACM MM 2025, SuSS 2025, ACL 2026, ICLR 2026, IEEE TAC, EMNLP 2026
Outreach: Association for the Advancement of Affective Computing (AAAC) DEI Committee, 2025